

SCREEN FREE NAVIGATION

FOR 2-WHEELER RIDERS - A WEARABLE HAPTIC BAND

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01. PROBLEM STATEMENT

NAVIGATION FOR TWO WHEELERS

- Two-wheeler riders commonly depend on smartphone navigation.
- Frequently **checking the phone screen** for directions diverts the rider's attention from the road and increases accident risk.
- With the rapid growth of **e-commerce and food delivery services**, the number of delivery riders on roads has significantly increased.
- These riders must navigate continuously between multiple locations under time pressure, making frequent phone interaction unsafe and inefficient.

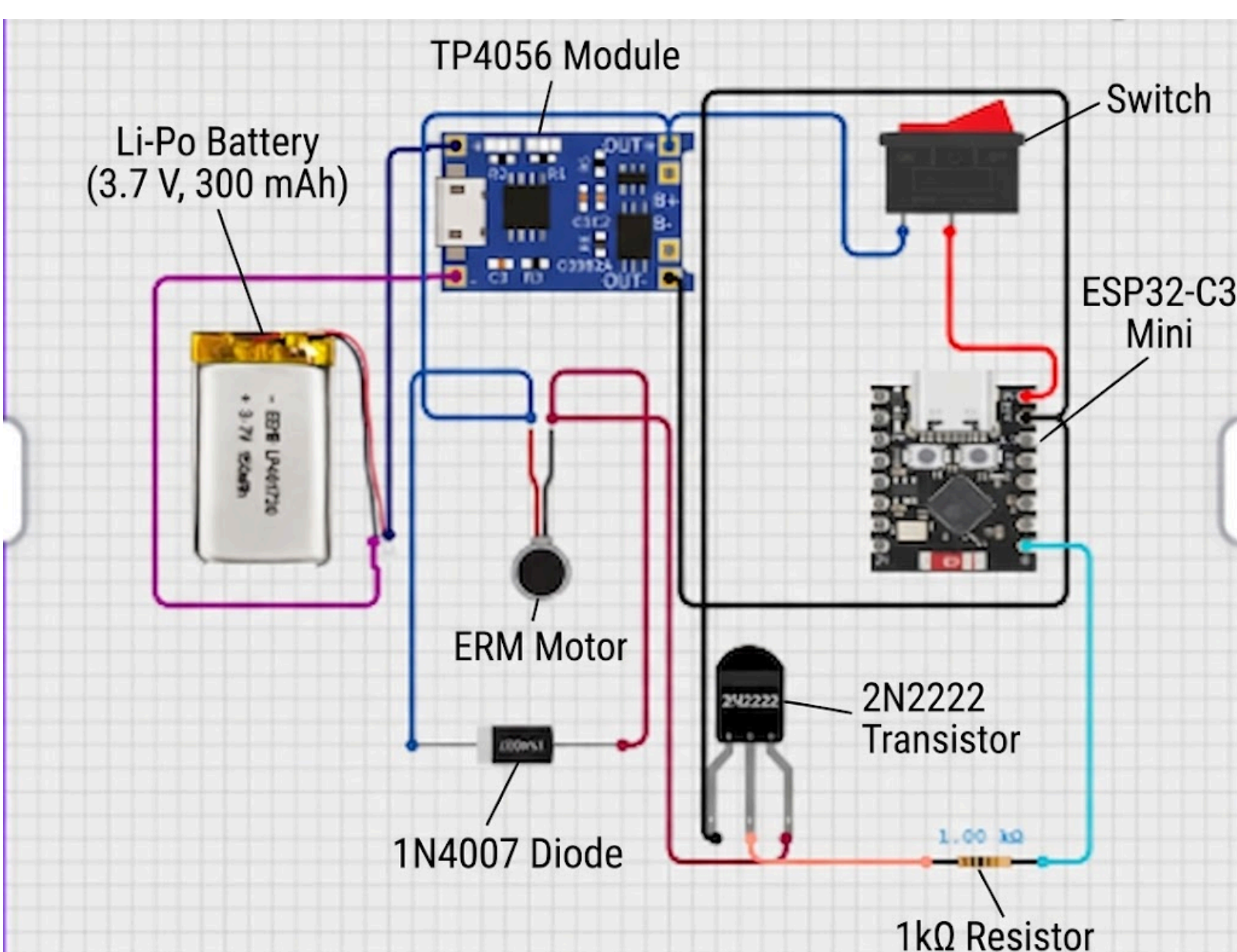
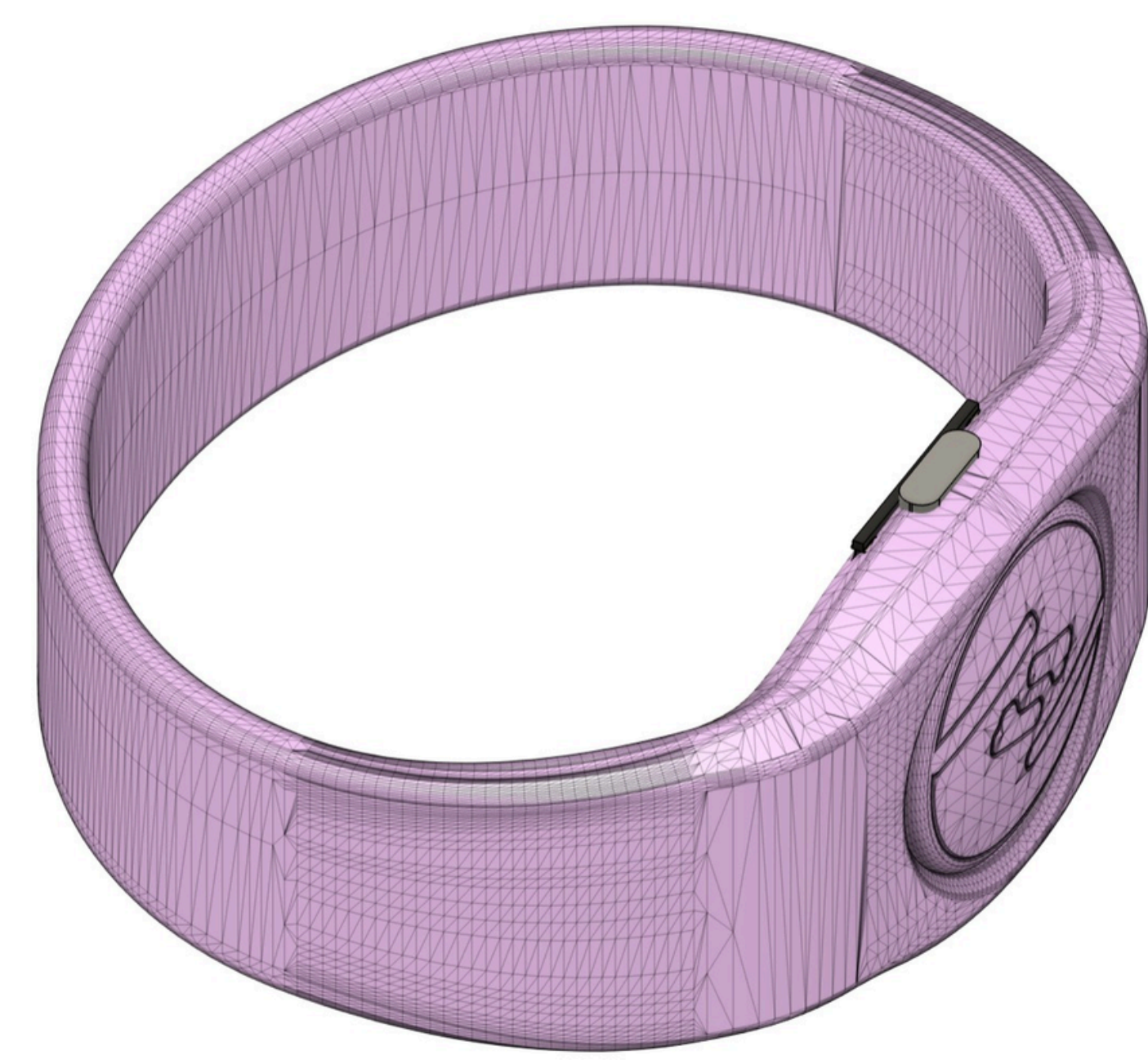
02. OBJECTIVES

- Reduce rider distraction caused by checking smartphone screens during navigation.
- Develop a screen-free navigation system using haptic feedback.
- Integrate smartphone GPS navigation with routing services.
- Provide clear directional instructions (left, right, straight) through vibration patterns.
- Improve rider safety by allowing riders to keep their eyes on the road.

03. REQUIREMENTS

- Riders carry a smartphone capable of GPS navigation.
- Bluetooth communication between smartphone and wearable hardware is available.
- Riders can interpret vibration patterns as directional signals.

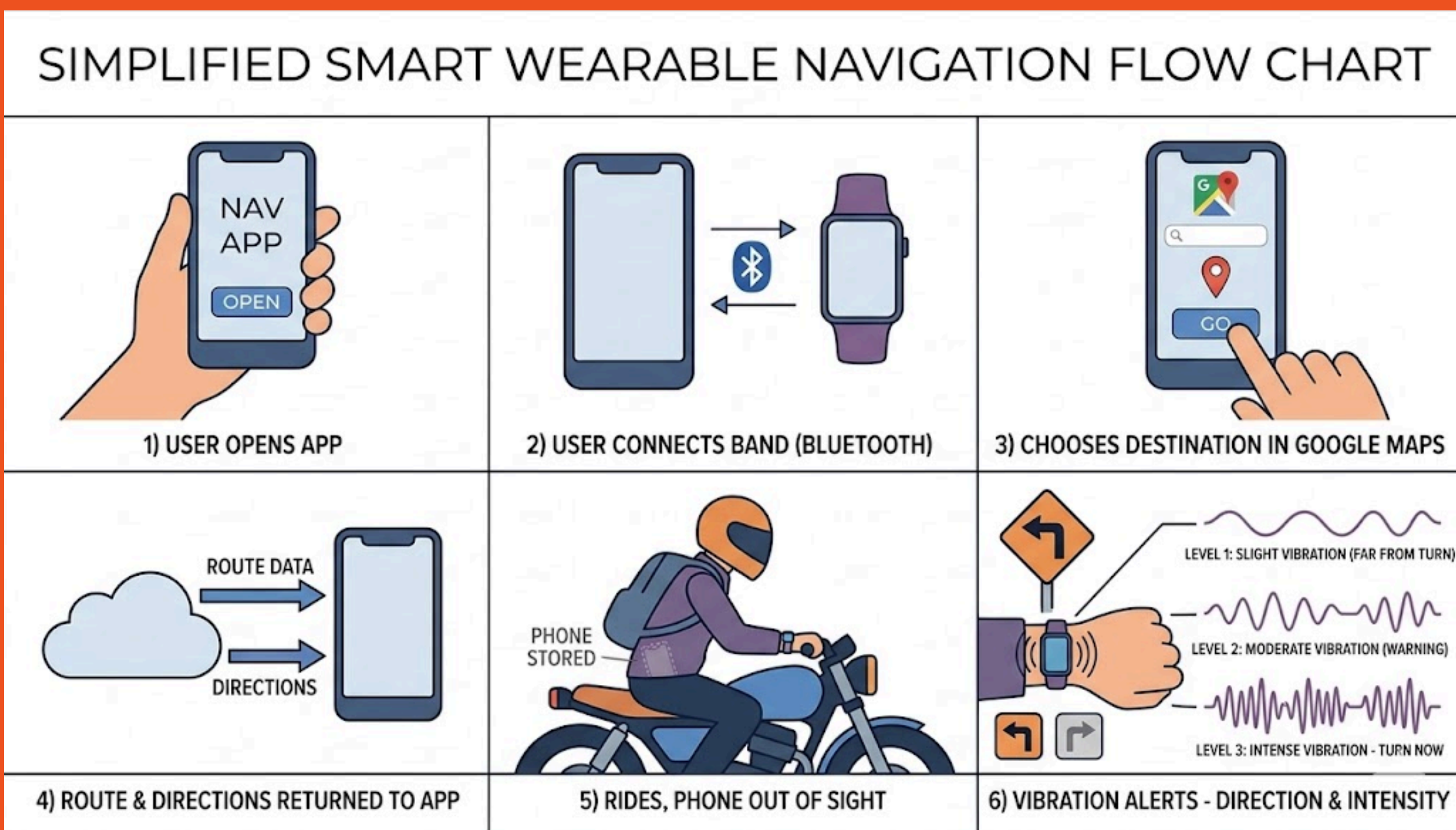
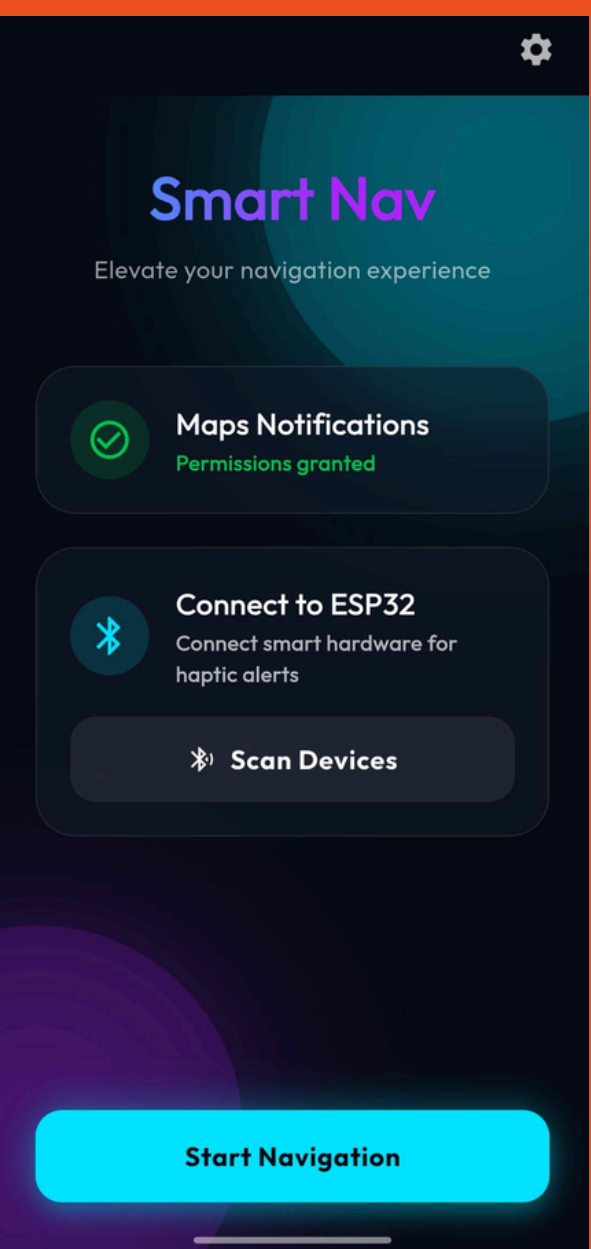
04. CONCEPT DESIGN



05. BILL OF MATERIALS

SNo.	Component	Qty	Cost
1	ESP32-C3 PRO Mini	2	₹628
2	ERM Coin Vibration Motor (8 x 3.4 mm)	2	₹122
3	3.7 V 300 mAh Li-Po Battery	2	₹270
4	TP4056 Li-ion Charging Module	2	₹28
5	Switches	6	₹28
6	1N4007 Diode	10	₹20
7	1kΩ Metal Film Resistor (MFR-25FR)	10	₹18
8	2N2222 NPN Transistor	5	₹10
Total			₹1124

06. PROTOTYPE



07. OUTCOME & INFERENCES

08. REFERENCES

- Kiss, F., Müller, J., Funk, M., & Alt, F. (2018). Navigation Systems for Motorcyclists: Exploring the Use of Wearable Tactile Feedback. CHI Conference on Human Factors in Computing Systems.
- Asif, M., Muthukumarasamy, V., & Elangovan, S. (2010). A Haptic Feedback-Based Navigation System for Mobile Devices. IEEE ICCAIE.

